

SAN MARTIN, Argentina

WMO: 874160

Lat: 33.0733S

Lon: 68.4650W

Elev: 653

StdP: 93.72

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.6	0.8	-8.7	1.9	7.9	-6.9	2.3	7.5	9.2	11.8	7.4	9.7	0.8	230	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	35.1	20.9	33.7	20.5	32.4	20.2	23.2	31.7	22.3	30.8	21.5	29.7	2.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	16.5	27.9	19.6	15.5	27.1	18.7	14.6	26.2	72.9	31.6	69.1	30.8	66.0	29.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.2	6.6	-2.8	38.3	1.3	1.6	-3.7	39.4	-4.5	40.3	-5.3	41.2	-6.2	42.3		
(5)			-4.1	25.3	1.2	1.2	-5.0	26.2	-5.7	26.8	-6.3	27.5	-7.2	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	25.9	24.3	21.4	16.9	12.9	9.5	8.9	11.5	14.8	18.9	22.1	24.7
	DBStd	6.73	2.98	3.31	3.32	3.44	3.15	2.58	3.04	3.69	3.91	3.76	3.50	3.27
	HDD10.0	139	0	0	0	2	11	37	57	25	7	1	0	0
	HDD18.3	1186	1	3	11	66	171	263	293	213	117	38	9	2
	CDD10.0	2919	492	399	353	209	99	24	22	73	151	277	363	457
	CDD18.3	924	234	169	106	23	1	0	0	3	11	56	122	201
	CDH23.3	8490	2390	1509	748	132	11	0	1	32	103	458	1052	2054
CDH26.7	3460	1110	623	236	18	1	0	0	6	19	133	393	922	
Wind	WSAvg	1.8	2.0	1.9	1.7	1.5	1.3	1.3	1.5	1.8	2.0	2.2	2.3	2.1
Precipitation	PrecAvg	168	33	38	26	8	4	4	3	2	9	6	17	27
	PrecMax	292	117	104	84	30	28	42	19	21	49	30	58	85
	PrecMin	82	3	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	61	24	33	22	9	7	8	5	4	12	8	17	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	35.7	32.9	28.8	24.8	20.3	21.7	27.2	29.0	32.6	34.7	36.7
		MCWB	21.6	22.3	21.2	18.6	16.1	12.1	11.7	15.0	16.4	17.8	19.1	20.6
	2%	DB	35.1	33.7	31.1	26.6	22.4	18.2	18.6	23.1	26.1	29.9	32.6	34.7
		MCWB	21.4	21.2	20.6	17.2	14.9	11.0	10.4	13.0	14.5	17.1	18.6	20.3
	5%	DB	33.6	32.1	29.3	24.6	20.2	16.6	16.6	20.5	24.0	27.9	30.7	33.3
		MCWB	21.0	20.9	19.9	16.5	13.5	10.1	9.2	11.6	13.6	16.1	18.0	20.0
	10%	DB	32.0	30.6	27.5	22.7	18.3	14.8	14.6	18.2	21.7	25.9	29.0	31.5
		MCWB	20.6	20.4	19.2	15.8	12.5	9.2	8.1	10.2	12.4	15.2	17.4	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	24.3	23.5	20.7	17.6	13.4	13.2	16.6	17.8	20.0	21.2	23.4
		MCDB	33.8	31.6	30.7	26.0	23.1	18.5	19.8	24.7	26.3	29.2	30.6	32.9
	2%	WB	23.3	23.0	21.9	18.7	15.8	11.8	11.0	13.7	15.5	18.5	20.0	22.1
		MCDB	32.1	31.1	28.5	23.8	20.6	16.7	17.1	22.0	24.5	27.2	29.8	32.0
	5%	WB	22.3	22.0	20.9	17.6	14.5	10.7	9.7	12.0	14.2	17.0	19.0	21.1
		MCDB	31.2	29.9	27.3	22.9	18.9	15.5	15.6	19.7	22.7	25.8	28.8	30.6
	10%	WB	21.4	21.2	19.9	16.5	13.3	9.6	8.5	10.7	13.0	15.9	18.1	20.2
		MCDB	30.2	28.5	26.3	21.6	17.4	14.0	13.9	17.5	20.5	24.4	27.3	29.5
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.0	10.6	10.5	10.4	11.2	11.7	12.0	11.6	11.8	11.9	11.9
		MCDBR	13.6	13.5	13.2	13.2	13.0	13.5	14.1	15.1	15.0	14.7	14.7	14.4
		MCWBR	4.9	4.6	5.4	6.4	7.0	8.0	8.0	8.1	7.2	6.1	5.7	5.3
	5% WB	MCDBR	12.0	11.4	11.3	11.5	11.5	12.3	13.3	14.7	13.8	13.0	13.1	12.6
MCWBR		5.5	4.9	5.2	6.1	6.7	7.8	7.9	8.3	7.1	6.0	5.7	5.5	
Clear-Sky Solar Irradiance	taub	0.394	0.388	0.368	0.362	0.325	0.302	0.298	0.323	0.364	0.366	0.374	0.389	
	taud	2.363	2.387	2.441	2.416	2.478	2.550	2.514	2.432	2.315	2.344	2.347	2.345	
	Ebn at Noon	944	929	912	858	843	845	867	887	900	940	957	953	
	Edh at Noon	131	123	109	100	83	73	79	96	122	127	132	134	
All-Sky Solar Radiation	RadAvg	7.87	6.86	5.63	4.29	3.10	2.85	3.12	4.05	5.32	6.66	7.83	8.27	
	RadStd	0.43	0.43	0.38	0.41	0.36	0.22	0.21	0.27	0.37	0.57	0.47	0.44	

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+0.48	+0.41	N/A	N/A	N/A	+63

Nomenclature: See separate page